

CO1-ASSESSMENT TOOL-2

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COURSE NAME: Deep Learning

COURSE CODE: MLAO4

COURSE OUTCOME COVERED

CO1: Compare and assess emerging technologies including AI/ML and Blockchain with respect to their applicability, societal impact, ethical considerations, and sustainability in the context of modern software system design and development.

ANNEXURE A

ERROR ANALYSIS TASK

Code of Conduct:

I ____K.V.SAI KRISHNA/192325037____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ANNEXURE A

A CNN model was trained to classify images into four categories: Cats, Dogs, Birds, and Rabbits. The model was tested on 200 images, producing the following confusion matrix.

Actual \ Predicted	Cat	Dog	Bird	Rabbit
Cat	42	5	2	1
Dog	6	38	3	3
Bird	1	4	43	2
Rabbit	2	3	5	40

Answer the following questions:

1. Calculate the overall accuracy of the model.
2. Identify which class has the highest number of misclassifications.
3. List the two most common classification errors made by the model.
4. Suggest three possible reasons why these errors occurred.
5. Recommend four techniques to improve the model performance.
6. Explain whether collecting more training data or increasing the number of CNN layers would be more beneficial in this case. Justify your answer.
7. If the model achieves 99% training accuracy but only 82% testing accuracy, what problem does the model suffer from? Explain how you would address it.

1. Overall Accuracy of the Model

The overall accuracy is calculated by dividing the total number of correct predictions (sum of the diagonal elements) by the total number of test samples (200).

$$\text{Accuracy} = \frac{42 + 38 + 43 + 40}{200} = \frac{163}{200} = 0.815 \text{ or } 81.5\%$$

2. Class with the Highest Number of Misclassifications

To find the class with the highest number of misclassifications, we sum the off-diagonal errors for each actual class (row total minus the correct diagonal value):

- **Cat:** $5 + 2 + 1 = 8$ errors (out of 50)
- **Dog:** $6 + 3 + 3 = 12$ errors (out of 50)
- **Bird:** $1 + 4 + 2 = 7$ errors (out of 50)
- **Rabbit:** $2 + 3 + 5 = 10$ errors (out of 50)

The **Dog** class has the highest number of misclassifications (12 errors).

3. Two Most Common Classification Errors

Looking at the off-diagonal values of the confusion matrix (Actual → Predicted):

1. **Dog misclassified as Cat:** 6 instances
2. **Cat misclassified as Dog:** 5 instances (tied with **Rabbit misclassified as Bird** at 5 instances)

The two most prominent confusion pairs involve mutual misclassifications between **Cats and Dogs** (6 and 5 errors), followed closely by **Rabbits predicted as Birds** (5 errors).

4. Three Possible Reasons Why These Errors Occurred

1. **Visual Similarity and Shared Features:** Cats and dogs share common pet-like features (fur, four legs, similar ear/eye configurations), leading to feature overlap in lower- and mid-level convolutional filters. Similarly, small furry animals like rabbits can share silhouettes with birds depending on posture, angle, or occlusion.
2. **Background and Context Clues:** Convolutional Neural Networks (CNNs) often rely heavily on background context (e.g., grass, indoor living rooms, cages). If an animal appears in an atypical environment, the model may misclassify it based on the background rather than the subject.
3. **Class Overlap and Ambiguity in Dataset:** Low-resolution images, poor lighting, or unusual camera angles can obscure key discriminative features (such as snout shape, ear length, or beak presence).

5. Four Techniques to Improve Model Performance

1. **Data Augmentation:** Apply geometric and photometric transformations (random rotations, horizontal flips, zooming, brightness/contrast adjustments, and cropping) to help the model learn invariant features and reduce overfitting.

2. **Transfer Learning:** Utilize a pre-trained backbone (such as ResNet, EfficientNet, or ConvNeXt pre-trained on ImageNet) and fine-tune it on this specific 4-class dataset. Pre-trained models already possess robust low- and mid-level feature extractors.
3. **Learning Rate Scheduling & Optimization:** Implement advanced optimizers (like AdamW) combined with a learning rate scheduler (such as Cosine Annealing) to help the network escape local minima and converge to a more optimal set of weights.
4. **Attention Mechanisms / Higher Resolution:** Incorporate channel/spatial attention modules (like SE-Blocks) to help the network focus on discriminative regions (e.g., snout vs. ears), and train on slightly higher input resolutions if details are currently being lost.

6. Collecting More Training Data vs. Increasing CNN Layers

Collecting more training data would be significantly more beneficial in this scenario.

- **Justification:** An accuracy of 81.5% on a relatively simple 4-class classification problem usually indicates that the model has learned the core concepts but lacks exposure to enough diverse variations (poses, lighting, breeds, backgrounds) to generalize robustly. Simply adding more CNN layers risks increasing model capacity unnecessarily, which can lead to severe overfitting, longer training times, and vanishing gradients if proper residual connections or normalization layers are absent. Providing a richer, more diverse dataset directly addresses feature variance and improves generalization.

7. Problem Identification and Remediation

Problem Identification

The model suffers from **overfitting** (high variance). This occurs when a model learns the training data *too well*—memorizing specific noise, quirks, and outliers in the training set—at the expense of learning generalized patterns, leading to poor performance on unseen test data.